ChatGPT said:

Yes. Since you're writing tomorrow, I'll make them **exam-style**, with **clear answers you can memorise directly**.

1. What is Agentic AI?

Answer:

Agentic AI is an AI system that can **plan, make decisions, use tools and take actions to achieve a goal with some level of independence**.

Cram:
Plan + Decide + Use tools + Act + Goal

---

2. What is an AI agent?

Answer:

An AI agent is a system that **receives information, makes decisions and performs actions to achieve a goal**.

Cram:
Input → Decide → Act → Goal

---

3. What is autonomy in Agentic AI?

Answer:

Autonomy means **how much freedom an AI system has to make decisions and take actions without a human controlling every step**.

---

4. What is the difference between an AI assistant and an AI agent?

Answer:

An AI assistant mainly **responds to user instructions**, while an AI agent can **plan, make decisions, use tools and work towards a goal more independently**.

Cram:

> Assistant = responds

> Agent = plans and acts

---

5. What are the five levels of autonomy?

Your notes describe five levels from rule-based automation to fully autonomous systems.

Unit 3A - AI Agents 101-Introdu…

Answer:

1. **Level 1 – Rule-based automation:** follows fixed rules.
2. **Level 2 – ML-assisted automation:** machine learning helps with specific tasks.
3. **Level 3 – Partial automation:** the agent can create and execute plans.
4. **Level 4 – High automation:** the agent works mostly independently and learns from experience.
5. **Level 5 – Fully autonomous:** the agent operates without human intervention.

Cram:

> Rules → ML → Plans → Learns → Fully autonomous

---

6. What are the main components of an AI agent?

Answer:

The main components are:

- **LLM/Model** – understands and reasons.
- **Goal** – tells the agent what to achieve.
- **Planning** – creates steps to achieve the goal.
- **Memory** – stores previous information.
- **Tools** – perform actions.
- **Knowledge** – provides useful information.
- **Orchestration** – controls how everything works together.

Cram:

> Model + Goal + Plan + Memory + Tools + Knowledge

---

7. What is the role of the LLM in an AI agent?

Answer:

The LLM is responsible for **understanding information, reasoning, planning and deciding what should happen next**.

Cram:

> **LLM = thinks and decides**

---

8. What is the role of tools in Agentic AI?

Answer:

Tools allow the agent to **perform real actions**, such as searching information, reading databases, doing calculations or sending messages.

Cram:

> **Tools = perform actions**

---

9. What is the difference between an LLM and a tool?

Answer:

The **LLM reasons and decides what should be done**, while the **tool performs the actual action**.

Example:

> The LLM decides that an email should be sent, while the email tool sends it.

Cram this:

> **LLM thinks, tools act.**

---

10. What is memory in Agentic AI?

Answer:

Memory allows the agent to **store previous information, actions and results so that it can maintain context and make better future decisions**.

Cram:

> Memory = remember the past to improve future decisions.

---

11. What is short-term memory?

Answer:

Short-term memory stores information needed for the **current conversation or task**.

Your notes explain that old information may need to be filtered or summarised because context windows are limited. Unit 3A - AI Agents 101-Introdu…

---

12. What is long-term memory?

Answer:

Long-term memory stores information that can be used **later or across different tasks**.

---

13. What are the three types of long-term memory?

Your notes identify semantic, procedural and episodic memory. Unit 3A - AI Agents 101-Introdu…

Semantic memory

Stores **facts and knowledge**.

Procedural memory

Stores **how to perform tasks or actions**.

Episodic memory

Stores **past experiences and outcomes**.

Cram:

> Semantic = facts
> Procedural = how
> Episodic = experiences

---

14. What is planning in Agentic AI?

Answer:

Planning is when the agent **breaks a goal into smaller steps and decides how to complete those steps**.

Cram:

> Planning = Goal → Steps

---

15. What is reasoning?

Answer:

Reasoning is the process of **analysing information and deciding the best action or answer**.

---

16. What is reflection?

Answer:

Reflection is when the agent **checks its own work, identifies mistakes and improves its answer or plan**.

Cram:

> Check → Fix → Improve

---

17. What is tool calling?

Answer:

Tool calling is when an agent **selects and uses an external tool to complete a task**.

Example:

> The agent can use a calculator, API, search tool or database.

---

18. What is collaboration in Agentic AI?

Answer:

Collaboration is when **multiple agents, or humans and agents, work together to complete a task**.

19. Name five Agentic AI patterns.

Your notes list these patterns directly. Unit 3A - AI Agents 101-Introdu…

Answer:

1. Planning/Reasoning
2. Reflection
3. Tool Calling
4. Memory
5. Collaboration

Cram:

> Plan → Reflect → Tools → Memory → Collaborate

---

20. What is ReAct?

Answer:

ReAct is an approach where an agent combines **reasoning and action**.

The agent:

> **Reasons → Acts → Observes → Reasons again**

Cram:

> ReAct = Reason + Act

---

21. What is prompt engineering?

Answer:

Prompt engineering is the process of **designing clear instructions and context so that an AI model produces a better response**.

Cram:

> Better prompt = better response.

---

22. What are the four main parts of a good prompt?

The notes give four parts: Instructions, Context, Input Data and Output Indicator.

Unit 3A - AI Agents 101-Introdu…

Instructions

Tell the model **what to do**.

Context

Give background information.

Input Data

Give the information to be analysed.

Output Indicator

Tell the model how the answer should be presented.

Cram:

> **Instructions + Context + Input + Output**

---

23. Why is context important in a prompt?

Answer:

Context helps the AI **understand the situation and produce a more relevant answer**.

---

24. Why are clear instructions important?

Answer:

Clear instructions tell the model exactly what to do and **reduce confusion and mistakes**.

---

25. Why is output format important?

Answer:

Output format tells the AI **how the final answer should look**, for example as a table, JSON, CSV or paragraph.

---

26. Create a simple prompt for a marketing AI agent.

Answer:

> **Role:** You are a marketing analyst.
> **Context:** The company wants to improve customer engagement.
> **Task:** Analyse customer feedback.
> **Input:** Customer feedback data.
> **Constraint:** Use only the provided data.
> **Output:** Give three problems and three recommendations in a table.

You can reuse this structure for almost any case study.

---

27. What is the System role?

Answer:

The System role gives the AI its **main instructions, rules or role**.

---

28. What is the User role?

Answer:

The User role contains the **request or question from the user**.

---

29. What is the Tool role?

Answer:

The Tool role contains **information returned by an external tool**.

---

30. What is the Assistant role?

Answer:

The Assistant role contains the **response generated by the AI**.

Your notes describe these four roles directly. Unit 3A - AI Agents 101-Introdu…

Cram:

> System = rules
> User = question
> Tool = external result
> Assistant = answer

---

31. What does RAG stand for?

Answer:

> **Retrieval-Augmented Generation**

---

32. What is RAG?

Answer:

RAG allows an AI system to **retrieve relevant information from an external knowledge source before generating an answer**.

Cram:

> Retrieve → Add context → Generate

Your notes show retrieved knowledge being added to the model context before generation.

Unit 3A - AI Agents 101-Introdu…

---

33. Why is RAG useful?

Answer:

RAG is useful because it:

- gives the AI relevant information,
- can improve answer accuracy,
- allows the AI to use external knowledge.

---

34. Explain the simple flow of RAG.

Answer:

> User asks a question → system retrieves relevant information → information is added to the prompt → LLM generates an answer.

Cram:

> Question → Retrieve → Context → LLM → Answer

---

35. What is an embedding?

Answer:

An embedding converts text or information into **numerical vectors so that similar information can be found**.

Cram:

> Embedding = text into numbers.

---

36. What is a vector database?

Answer:

A vector database stores embeddings and helps the system **find information similar to the user's question**.

---

37. What is an Agentic AI framework?

Answer:

An Agentic AI framework provides a **structure for building and managing agents, tools, tasks, models and workflows**.

---

38. What is orchestration?

Answer:

Orchestration is the process of **coordinating the model, memory, tools and tasks so that they work together correctly**.

Cram:

> Orchestration = manages the whole agent.

---

39. What is state in an AI agent?

Answer:

State represents the agent's **current information and progress while completing a task**.

---

40. Why are Agentic AI frameworks useful?

Answer:

They make it easier to:

- connect different models,
- manage tools,
- control workflows,
- manage tasks and state,
- support multiple agents.

---

41. What factors should be considered when choosing an LLM?

Your notes list model characteristics such as modality, architecture, training data, fine-tuning, benchmarks, model size, reasoning and safety data. Unit 3A - AI Agents 101-Introdu…

Answer:

Consider:

- task suitability,
- reasoning ability,
- model size,
- safety,
- training data,
- accuracy/benchmarks,
- modality,

- cost.

Cram:

> Task + Reasoning + Accuracy + Size + Safety + Cost

---

42. Why can different models be used for different tasks?

Answer:

Different models have different strengths.

For example:

> one may be better at reasoning, another at summarisation, and another at specialised tasks.

---

43. What is a model card?

Answer:

A model card is a document that describes a model's **purpose, capabilities, limitations, training information and safety information**.

---

44. What are the main challenges of Agentic AI?

Your notes list safety and trust, bias, copyright/IP, governance/ethics, hallucinations, relevance, data security and privacy. Unit 3A - AI Agents 101-Introdu…

Answer:

Main challenges include:

- safety,
- trust,
- bias,
- hallucinations,
- privacy,
- security,
- ethics,
- governance.

Cram:

> Safety + Bias + Hallucination + Privacy + Ethics

---

45. What is hallucination?

Answer:

Hallucination is when an AI generates **false or incorrect information as if it were true**.

Cram:

> Hallucination = confident but wrong.

---

46. How can hallucinations be reduced?

Answer:

Hallucinations can be reduced by:

- using reliable data,
- using RAG,
- giving clear prompts,
- checking the output,
- using human review.

Cram:

> Reliable data + RAG + Verify + Human

---

47. What is bias in AI?

Answer:

Bias happens when an AI system produces **unfair results for certain people or groups**.

---

48. How can bias be reduced?

Answer:

Bias can be reduced by:

- using representative data,
- testing for unfair results,
- using human oversight.

---

49. Why is privacy important in Agentic AI?

Answer:

Privacy is important because agents may access sensitive information, so data must be **protected and only used when necessary**.

---

50. What is Human-in-the-Loop?

Answer:

Human-in-the-Loop means that a **human reviews or approves important AI decisions or actions**.

Your notes show HITL approval being used to improve safety and reliable operation.

Unit 3A - AI Agents 101-Introdu…

Cram:

> HITL = human checks AI.

---

51. Why is Human-in-the-Loop important?

Answer:

It improves:

- safety,
- accountability,
- reliability,
- human control.

---

52. Traditional AI vs Generative AI vs Agentic AI

Answer:

Traditional AI

Predicts, classifies or follows rules.

Generative AI

Creates new content.

Agentic AI

Plans and takes actions towards a goal.

Cram:

> Traditional = Predict
> GenAI = Create
> Agentic = Act

---

53. Workflow vs Agent

Answer:

A workflow follows **predefined steps**.

An agent can **decide which steps to take depending on the situation**.

Cram:

> Workflow = fixed steps
> Agent = chooses steps

---

54. Memory vs Knowledge

Answer:

Memory stores information from **past interactions and actions**.

Knowledge is information available from **documents, databases or other sources**.

Cram:

> Memory = past
> Knowledge = available information

---

55. Planning vs Reflection

Answer:

Planning decides **what should be done**.

Reflection checks **what was done and how it can be improved**.

Cram:

> Planning = before
> Reflection = after

---

56. Case study: A university wants an AI agent to recommend modules. Identify the goal, task, tool and memory.

Answer:

Goal: Recommend suitable modules.

Task: Analyse the student's academic record.

Tool: Student database.

Memory: Previous student choices and recommendations.

---

57. Case study: A marketing agent must create a campaign. How can it work?

Answer:

The agent can:

1. receive the goal,
2. analyse customer information,
3. create a plan,
4. use tools,
5. generate campaign ideas,
6. check the results,
7. improve the plan.

Cram:

> Goal → Plan → Tool → Act → Check → Improve

---

58. What safeguards are needed for a high-risk AI decision?

Answer:

Use:

- human approval,
- privacy protection,
- bias testing,
- accuracy checks.

Cram:

> Human + Privacy + Bias + Accuracy

---

59. What should remain a human responsibility?

Answer:

Humans should make final decisions for **high-risk, ethical, legal or sensitive matters**.

Examples:

- hiring,
- firing,
- medical decisions,
- important financial decisions.

---

60. What tasks can be delegated to AI agents?

Answer:

AI agents can handle lower-risk tasks such as:

- searching information,
- analysing data,
- creating drafts,
- summarising information,

- making recommendations.

THE 10 ANSWERS I WANT YOU TO CRAM FIRST

If you cannot learn all 60 tonight, learn these **10 first**:

1. **Agentic AI can plan, decide, use tools and act towards a goal.**
2. **Autonomy is how much freedom AI has to make decisions without human help at every step.**
3. **Assistant responds; agent plans and acts.**
4. **LLM thinks, tools act, memory remembers.**
5. **Planning breaks a goal into smaller steps.**
6. **Prompt = Instructions + Context + Input + Output.**
7. **RAG = Retrieve → Add context → Generate.**
8. **ReAct = Reason → Act → Observe → Reason again.**
9. **Reflection = Check → Fix → Improve.**
10. **Traditional AI predicts, GenAI creates, Agentic AI acts.**

And for a longer answer, memorise this:

> **An AI agent uses an LLM for reasoning and planning, memory for storing previous information, knowledge for context and tools for performing actions so that it can work towards a goal with some level of autonomy.**